

Validation State Transition & Revalidation Trigger Module (VSTRM)

Architecture Ecosystem: Structured Reality Standards™
Architecture Family: VALIDOS® — Validation Governance Architecture
Parent Standard: Validation State Governance Standard (VSGS)
Operational Layer: Validation State Governance Layer
Category: AI & Interpretation
Subcategory: Validation Governance
Type: Parent Standard Module
Governed Space: Validation State Transition & Revalidation Triggering
Version: 1.0
Status: Canonical · Open Module
Effective Date: 12 August 2026
Compatibility: OOF® Methodology OS™ · GOA™ · OBIDENITY® · INTEGROS®
· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Governed Space

Validation State Transition & Revalidation Triggering

Canonical Definition

Validation State Transition & Revalidation Trigger Module (VSTRM)

defines the governance framework for determining when an active Validation State should remain, restrict, become conditional, suspend, expire, invalidate, enter conflict, become partial, or transition into Revalidation Required based upon material change, applicability loss, expiration, determination change, new evidence, incidents, uncertainty, conflict, drift, and other validation-relevant lifecycle events.

It establishes the decision layer through which detected change becomes a governed Validation State transition or a formal Revalidation Trigger.

Operational Role

VSTRM serves as the fourth internal governance space of the Validation State Governance Standard.

The preceding modules establish:

VSAEM — Which Validation State may legitimately be assigned?

VSCAM — Where, when, and under which conditions does that state apply?

VMCSMM — What has changed, and could that change materially affect the state?

VSTRM asks:

Given the detected change or lifecycle event, what should happen to the current Validation State, and is revalidation now required?

The governed progression is:

Current Validation State → State Impact Signal → Transition Eligibility Assessment → Transition Classification → Revalidation Trigger Assessment → Transition Authorization → New Validation State / Revalidation Required

VSTRM does not perform revalidation itself.

It determines when the existing Validation State can remain and when a new validation cycle becomes necessary.

Module Operational Space

VSTRM governs:

- Validation State transition,
- transition eligibility,
- state maintenance,
- state restriction,
- state conditionalization,
- partial-state transition,
- state suspension,
- state resumption,
- state expiration,
- state invalidation,
- state conflict,
- Revalidation Required,
- revalidation triggers,
- material-change triggers,
- critical-change triggers,
- applicability-loss triggers,
- condition-failure triggers,
- incident triggers,
- evidence-change triggers,
- source-change triggers,
- determination-change triggers,
- risk-change triggers,
- drift triggers,
- cumulative-change triggers,
- time-based triggers,

- event-based triggers,
 - transition authority,
 - transition precedence,
 - transition constraints,
 - transition reversibility,
 - transition history,
 - and transition traceability.
-

Module Function

VSTRM applies when a Validation State Impact Signal, lifecycle event, expiration condition, or other validation-relevant change requires a decision about the current state.

Its function is to prevent:

- active validation states persisting despite known loss of applicability,
 - minor changes automatically triggering unnecessary full revalidation,
 - critical changes being treated as ordinary maintenance,
 - expired validation remaining represented as current,
 - suspended states resuming automatically without reassessment,
 - negative new evidence being ignored,
 - positive historical determination overriding current state reality,
 - revalidation triggers remaining undefined,
 - state transitions occurring informally,
 - and state changes being made without traceable methodological basis.
-

State Transition

A **Validation State Transition** is a governed change from one formal Validation State to another.

Examples include:

Validated → Conditionally Validated

Validated → Validation Suspended

Validated → Validation Expired

Validated → Revalidation Required

Conditionally Validated → Validated

Conditionally Validated → Validation Suspended

Validation Suspended → Validated

Validation Suspended → Revalidation Required

Validation Pending → Validated

Validation Pending → Partially Validated

Validation Required → Validation Pending

Validated → Invalidated

A transition must have a governed trigger and basis.

State Maintenance

Not every detected change should produce transition.

VSTRM may determine:

State Maintained

where:

- change is immaterial,
 - applicability remains intact,
 - object equivalence remains sufficient,
 - conditions remain satisfied,
 - and no material lifecycle trigger requires a new state.
-

State Maintenance With Observation

Some minor changes may not justify transition but may require continued monitoring.

A state may therefore remain active with:

Enhanced Monitoring

or:

Targeted Review

without immediate revalidation.

State Restriction

A Validation State may need to remain active only within a narrower scope.

For example: **Validated for Environments A and B**

may become:

Validated for Environment A Only

after a material issue appears in Environment B.

Restriction preserves valid portions while removing unsupported applicability.

State Conditionalization

An unrestricted Validated state may transition into:

Conditionally Validated

where new information introduces a continuing condition necessary to preserve validity.

For example:

Validated → Conditionally Validated with Human Oversight

after a newly identified failure mode requires oversight as a mitigation.

Partial-State Transition

Where one portion of the Validation Object remains supported and another does not, VSTRM may transition:

Validated → Partially Validated

rather than invalidating the complete object.

This requires reliable separation of the affected scope.

State Suspension

A Validation State may be suspended where current applicability becomes materially uncertain but the validation basis has not yet been conclusively destroyed.

Possible triggers include:

- critical incident,
 - unresolved Material Change,
 - disputed source reliability,
 - new contradictory evidence,
 - determination suspension,
 - unknown object equivalence,
 - critical control loss,
 - or active investigation.
-

Suspension as a Protective State

Suspension is designed to preserve methodological caution.

It says:

The previous state may still prove legitimate, but current unrestricted applicability cannot presently be defended.

State Resumption

A suspended state may resume where sufficient basis exists to establish that:

- the triggering concern has been resolved,
- object equivalence remains,
- the applicable determination remains valid,
- required conditions are restored,
- and no unresolved material issue remains.

Resumption may return to:

Validated

Conditionally Validated

Partially Validated

or another appropriate state.

Resumption Is Not Automatic

VALIDOS® establishes:

Condition Restored ≠ Validation State Automatically Restored

The effect of the disruption must first be assessed.

State Expiration

A state transitions to:

Validation Expired

when its governed validity period, event-based validity condition, or expiration rule is reached.

Expiration may be:

- scheduled,
-

- event-triggered,
 - condition-triggered,
 - or determined by lifecycle rules.
-

Expiration Is Not Negative Validation

VALIDOS® establishes:

Validation Expired ≠ Negative Validation Determination

Expiration means the current validation basis is no longer sufficiently current.

The object may still be acceptable, but the architecture requires renewed validation basis.

State Invalidation

A state may transition to:

Invalidated where sufficient basis exists to determine that the previous state cannot legitimately remain.

Possible causes include:

- invalid underlying determination,
 - invalid critical execution,
 - major object change destroying equivalence,
 - critical evidence compromise,
 - decisive Source Reliability failure,
 - fundamental methodological defect,
 - or new evidence demonstrating that the state basis was materially wrong.
-

Invalidation Is Stronger Than Revalidation Required

VALIDOS® distinguishes:

Revalidation Required

from:

Invalidated

Revalidation Required means the existing basis is insufficient to support continued current validation.

Invalidated means the previous validation basis has been materially destroyed or shown to be illegitimate.

Validation Conflict Transition

A Validation Object may transition into:

Validation Conflicted

where new evidence, competing determinations, or materially incompatible current states prevent one coherent status.

This may occur before sufficient basis exists for suspension, invalidation, or a new positive state.

Revalidation Trigger

A **Revalidation Trigger** is a governed condition indicating that a new validation cycle is required.

A trigger may arise from:

- Material Change,
 - Critical Change,
 - expiration,
 - scope expansion,
 - new use case,
 - new population,
 - new environment,
 - new jurisdiction,
 - capability expansion,
 - autonomy escalation,
 - dependency replacement,
 - control removal,
 - serious incident,
 - new evidence,
 - Source Reliability change,
 - determination reassessment,
 - cumulative drift,
 - or regulatory requirement.
-

Trigger Does Not Mean Full Revalidation

VALIDOS® establishes:

Revalidation Trigger ≠ Automatic Full Revalidation

The required revalidation depth may be:

- targeted,
 - partial,
 - criterion-specific,
 - scope-specific,
 - change-specific,
 - subsystem-specific,
 - or complete.
-

The final Parent Standard will govern the full revalidation process.

Material Change Trigger

A confirmed Material Change may trigger reassessment.

Whether it triggers revalidation depends upon whether the existing validation basis can still legitimately represent the changed object.

Critical Change Trigger

A Critical Validation-Relevant Change may justify immediate:

- suspension,
- restriction,
- Revalidation Required,
- or other protective transition

before complete analysis is finished.

Applicability Loss Trigger

Where VSCAM establishes full or partial applicability loss, VSTRM determines whether the state should:

- restrict,
 - become partial,
 - suspend,
 - expire,
 - or enter Revalidation Required.
-

Condition Failure Trigger

Failure of a mandatory continuing condition may trigger immediate state change.

For example:

Conditionally Validated with Human Override Human Override Removed

may trigger:

Revalidation Required

or:

Validation Suspended

depending upon the validation basis and operational context.

Incident Trigger

A significant incident may require state reassessment even where no formal object change occurred.

Incident severity may influence whether the response is:

- monitoring,
 - restricted state,
 - suspension,
 - revalidation,
 - or invalidation.
-

Evidence Trigger

New evidence may alter current validation confidence.

Examples include:

- new independent test,
 - adverse-event evidence,
 - contradictory field data,
 - improved long-term data,
 - newly discovered edge-case behavior,
 - or evidence correcting a prior assumption.
-

Source Trigger

Changes in a decisive source may trigger State reassessment.

Examples include:

- source retraction,
 - calibration failure,
 - provenance failure,
 - compromised system,
 - newly discovered conflict,
 - or loss of capability.
-

Determination Trigger

If the Overall Validation Determination is:

- corrected,
- modified,
- suspended,
- withdrawn,
- or superseded,

the Validation State should be reassessed.

Risk Trigger

A change in consequence, exposure, or risk may change the required validation assurance even if the object remains unchanged.

For example:

A model previously used for low-consequence advice may later control a critical infrastructure process.

The same validation basis may no longer be sufficient.

Scope Expansion Trigger

Expansion into a new function, market, population, jurisdiction, environment, or autonomy level may create a new validation question.

The existing state should remain valid only within its original scope unless the expansion is independently supported.

Drift Trigger

Validation Drift may eventually reach a threshold where the current state no longer remains sufficiently representative.

Drift-triggered revalidation is particularly important for adaptive or continuously updated systems.

Cumulative Change Trigger

Several minor changes may collectively reach revalidation significance.

VSTRM therefore consumes the Cumulative Change assessment from VMCSMM.

Time-Based Revalidation Trigger

Some objects may require revalidation after a fixed period.

For example:

Revalidate every 12 months

where justified by the domain.

The period itself should have methodological rationale.

Event-Based Revalidation Trigger

Revalidation may occur when specific events happen rather than according to calendar time.

Examples include:

- model retraining,
 - safety architecture change,
 - ownership transfer affecting dependencies,
 - major infrastructure migration,
 - autonomy level increase,
 - or deployment to a new environment.
-

Condition-Based Revalidation Trigger

Failure or change of a continuing state condition may trigger revalidation.

Revalidation Trigger Materiality

Not every trigger should produce the same response.

VSTRM may classify triggers as:

Advisory Trigger

Review recommended.

Targeted Revalidation Trigger

Only specific affected validation domains require new validation.

Partial Revalidation Trigger

A defined portion of the object or scope requires revalidation.

Full Revalidation Trigger

The current validation basis is no longer sufficient across the intended scope.

Critical Revalidation Trigger

Immediate protective transition and high-priority revalidation are required.

Revalidation Trigger Scope

Every trigger should identify the affected scope.

For example:

Revalidation Required — Safety Domain Only

or:

Revalidation Required — Version 5.0 Complete System

This supports proportionate response.

Transition Eligibility Assessment

Before state transition, VSTRM evaluates whether sufficient governed basis exists for the proposed transition.

For example:

To transition:

Validation Suspended → Validated

the architecture should possess sufficient evidence that the suspension condition has been resolved.

Transition Basis

A formal transition basis may include:

- Current Validation State,
 - applicable Validation Determination,
 - State Applicability Profile,
 - Material Change Assessment,
 - State Impact Signal,
 - incident record,
 - evidence change,
 - source change,
 - risk change,
 - trigger,
 - and transition rationale.
-

Transition Precedence

Some transitions may take precedence over ordinary state continuity.

For example:

A Critical State Suspension Trigger may override an active Validated state immediately.

Historical validated status remains visible, but the current state becomes suspended.

Transition Constraints

Certain transitions may require prerequisites.

For example:

Validation Expired → Validated

may require an applicable renewed determination.

Invalidated → Validated

should ordinarily require a new legitimate validation basis rather than simple administrative restoration.

Transition Reversibility

Some transitions are reversible.

For example:

Validated → Suspended → Validated

Others may require new validation rather than reversal.

For example:

Validated → Invalidated

should not ordinarily return directly to Validated without a new validation basis.

Transition Authorization

Where formal authority is required, VSTRM may govern:

- transition authority,
 - suspension authority,
 - invalidation authority,
 - resumption authority,
 - revalidation-trigger authority,
 - escalation authority,
 - and delegated transition rights.
-

Automated State Transition

Some transitions may be automated where rules are explicit.

Examples include:

Expiration Date Reached → Validation Expired

or:

Critical Configuration Hash Change → Revalidation Required

Automation should remain traceable and subject to governed exception handling.

AI-Assisted Transition Assessment

AI may assist with:

- trigger classification,
- materiality synthesis,
- transition-path analysis,
- state comparison,
- scope localization,
- or revalidation recommendation.

AI must not silently invent state transitions outside the governed transition model.

Transition Conflict

A Transition Conflict exists where available signals support materially different state responses.

For example:

one governance pathway suggests:

Conditionally Validated

while another suggests:

Validation Suspended.

Such conflict may require escalation or temporary protective state.

Protective Transition

Where critical uncertainty exists, the architecture may permit a more conservative temporary transition while the final state is assessed.

For example: **Validated → Validation Suspended**

pending investigation.

This preserves methodological caution without prematurely invalidating the object.

State Transition Record

A formal transition record may preserve:

- Transition Identifier,
 - Validation Object,
 - previous state,
 - proposed state,
 - actual new state,
 - trigger,
 - affected scope,
 - materiality,
 - transition basis,
 - restrictions,
 - conditions,
 - authority,
 - effective point,
 - revalidation requirement,
 - and sufficient traceability.
-

Revalidation Trigger Record

A Revalidation Trigger Record may preserve:

- Trigger Identifier,
 - Validation Object,
 - current state,
 - trigger type,
 - triggering event,
 - affected scope,
 - affected criteria or domains,
 - trigger severity,
 - recommended revalidation depth,
 - timing requirement,
 - state effect,
 - and traceability references.
-

State Transition History

VSTRM preserves the lifecycle chain:

Previous State → Trigger → Assessment → Transition → New State

This prevents current state labels from losing historical context.

State Change Without Revalidation

Some transitions may occur without revalidation.

For example:

Validated → **Validation Expired** is a lifecycle transition.

Validated → **Validation Suspended**

may occur because of a critical investigation.

A new validation cycle may follow, but transition itself does not require completion of revalidation first.

Revalidation Without Complete State Loss

A state may remain partially valid while revalidation occurs for affected domains.

For example:

Performance: Validated

Safety: Revalidation Required

This enables parallel lifecycle governance.

Transition and Reliance

VSTRM does not govern final reliance decisions.

However, state transitions create critical inputs for the final VALIDOS® Parent Standard.

For example:

Validation Suspended

may materially restrict whether existing reliance can continue.

The tenth standard governs that relationship.

Minimum Implementation Framework

1. Receive State Impact Signal

Consume relevant change, applicability, expiration, incident, or determination signals.

2. Identify Current Validation State

Establish the state currently governing the object.

3. Assess Transition Eligibility

Determine which state transitions are methodologically available.

4. Assess Revalidation Trigger

Determine whether new validation is:

- unnecessary,
- advisory,
- targeted,
- partial,
- full,
- or critical.

5. Identify Affected Scope

Localize the transition and revalidation requirement.

6. Select Governed Transition

Determine whether the state should:

- remain,
- restrict,
- become conditional,
- become partial,
- suspend,
- resume,
- expire,
- enter conflict,
- invalidate,
- or become Revalidation Required.

7. Establish Transition Authority

Confirm required authority where applicable.

8. Establish Effective Point

Record when the new state becomes current.

9. Generate Revalidation Trigger Record

Where new validation is required, define the trigger and affected scope.

10. Preserve Lifecycle Traceability

Maintain:

- previous state,
- trigger,
- evidence,
- change assessment,
- applicability assessment,
- materiality,
- transition decision,
- new state,
- conditions,
- restrictions,
- affected scope,
- revalidation requirement,
- authority,
- effective point,
- and sufficient historical continuity.

Governance Outputs

VSTRM may produce:

- Validation State Transition Assessment,
- Validation State Transition,
- State Maintenance Record,
- State Restriction Record,
- State Conditionalization Record,
- Partial-State Transition Record,
- Validation Suspension Record,
- Validation Resumption Record,
- Validation Expiration Record,
- Validation Invalidation Record,
- Validation Conflict Transition Record,
- Revalidation Required Record,
- Validation Revalidation Trigger,
- Advisory Revalidation Trigger,
- Targeted Revalidation Trigger,
- Partial Revalidation Trigger,
- Full Revalidation Trigger,
- Critical Revalidation Trigger,
- Revalidation Trigger Scope Record,
- Revalidation Trigger Materiality Assessment,
- State Transition Constraint Record,
- State Transition Conflict Record,
- Protective Transition Record,
- State Transition Authority Record,
- State Transition Effective Point,
- Validation State Transition Record,
- Revalidation Trigger Record,
- and Validation State Transition History.

Use Case 1 — AI Model Capability Expansion

Scenario

An AI model was Validated for advisory output with no external tool access.

The provider enables autonomous tool execution.

Application

VMCSMM identifies a Critical Validation-Relevant Capability Change.

VSTRM assesses whether the prior validation basis covers autonomous tool use.

It does not.

Result

The object transitions:

Validated → **Revalidation Required**

for the newly autonomous scope.

The previously validated advisory scope may remain separately valid where methodologically separable.

Use Case 2 — Healthcare AI Incident

Scenario

A healthcare AI system is currently Validated.

A serious adverse event reveals a previously unobserved failure mode.

The causal relationship is not yet established.

Application VSTRM does not immediately declare the system Invalidated.

The new evidence creates material uncertainty requiring investigation.

Result

The state may transition:

Validated → **Validation Suspended**

pending reassessment.

If the event later proves unrelated, the state may resume.

If it exposes a fundamental validation failure, revalidation or

invalidation may follow.

Use Case 3 — Industrial System Expiration

Scenario

An industrial system's Validation State has a 12-month governed validity period.

The period expires without a renewed validation cycle.

Application

The expiration condition is unambiguous.

Result

The state transitions:

Validated → Validation Expired

and a Revalidation Trigger is generated.

The system is not labeled as having failed validation.

Architectural Position

Validation State Transition & Revalidation Trigger is the fourth internal governance space of the **Validation State Governance Standard (VSGS)**.

The progression now becomes:

State Assignment & Eligibility → State Conditions & Applicability → Material Change & State Monitoring → State Transition & Revalidation Triggering

VSAEM establishes:

Which state may the object receive?

VSCAM establishes:

Where and under which conditions does that state apply?

VMCSMM establishes:

What has changed and whether it may matter?

VSTRM establishes: **What happens to the state because of that change, and whether revalidation is now required?**

The fifth and final VSGS module can then govern:

How is the complete Validation State history preserved across assignments, transitions, suspensions, expirations, invalidations, supersession, and revalidation cycles so that the object's current and historical validation condition remains reconstructable?

The VSGS progression therefore continues:

State Assignment & Eligibility → State Conditions & Applicability → Material Change & State Monitoring → State Transition & Revalidation Triggering → State Continuity & Lifecycle Record

Foundational Principle

A Validation State should change when the governed basis supporting that state materially changes—not merely when an organization chooses to change the label.

Canonical Closing Statement

Validation State Transition & Revalidation Trigger Module establishes the lifecycle decision layer of Validation State Governance within VALIDOS® by governing how material change, applicability loss, condition failure, expiration, incidents, evidence and source changes, determination changes, risk shifts, drift, and cumulative change affect the current Validation State. Through state maintenance, restriction, conditionalization, suspension, resumption, expiration, invalidation, conflict, transition eligibility, transition authority, revalidation-trigger classification, scope localization, and traceability, VSTRM ensures that current validation status evolves in response to governed reality and that new validation is required precisely when the previous validation basis can no longer legitimately support the state it once established.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

[→ Previous module: Validation Material Change & State Monitoring Module \(VMCSMM\)](#)

[→ Next module: Validation State Continuity & Lifecycle Record Module \(VSCLRM\)](#)

Related Documents

[→ Validation State Governance Standard](#)

[→ About Validation State Governance Standard](#)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>